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ICG System Green Near-Infrared (NIR) Fluorescence Imaging System

Technical Specifications and Salient Characteristics

1. System Overview

The ICG System Green Fluorescence Imaging System is an advanced endoscopic visualization platform designed for real-time near-infrared (NIR) fluorescence imaging combined with high-resolution white light endoscopy. The system enables simultaneous visualization of anatomical structures and fluorescence-enhanced tissue perfusion using indocyanine green (ICG) dye.

- Intended clinical applications include:
 - Intraoperative tissue perfusion assessment
 - Fluorescent cholangiography
 - Lymphatic mapping and sentinel lymph node identification
 - Vascular flow assessment
 - Minimally invasive laparoscopic and thoracoscopic procedures

2. Salient Features

- Simultaneous white light and ICG/NIR fluorescence imaging capability
- Real-time fluorescence overlay onto standard endoscopic image
- Optional grayscale fluorescence mode for enhanced tissue contrast
- Native 4K Ultra High Definition (UHD) image output
- High-sensitivity dedicated NIR sensor
- LED-based illumination optimized for visible and NIR wavelengths
- Automatic light intensity regulation synchronized with camera controller
- Extended LED lifespan (up to 15,000 hours)
- Integrated fixed-focus lens (~24 mm focal length)
- Rigid telescopes available in 10 mm diameter with multiple viewing angles
- Fused-glass fiber light cable with enhanced light transmission
- Touchscreen user interface with upgradeable software capability
- Medical-grade 4K display compatibility
- Modular architecture for system expansion and upgrades

3. Technical Specifications

1. **Imaging Resolution:** Native 4K UHD
2. **Fluorescence Type:** ICG-based Near-Infrared (NIR)

3. **Excitation Range:** 600–900 nm
4. **Emission Detection Range:** 800–850 nm
5. **Imaging Modes:** White Light / Fluorescence Overlay / Fluorescence Pure
6. **Light Source Type:** LED-based
7. **LED Lifespan:** Up to 15,000 hours
8. **Telescope Diameter:** 10 mm
9. **Viewing Angles:** 0°, 30°, 50°
10. **Fiber Cable Diameter:** 5.0 mm
11. **Fiber Cable Length:** 3.0 m
12. **Connectivity:** HDMI, RJ45 network interface, USB
13. **Sterilization:** Autoclavable components where applicable

4. Typical System Configuration

- 4K Camera Controller Unit
- NIR-Optimized Camera Head with Integrated Lens
- LED Light Source for White Light and NIR Imaging
- Rigid Telescope (10 mm)
- Fusion Fiber Light Cable (5.0 mm)
- Medical-Grade 4K Monitor
- Remote Control
- Footswitch
- Power and Interface Cables
- Sterilizable Accessories and Mounting Hardware

5. Extra attachments

- Telescope 30° 10MM WL 305MM – rigid
- Universal Tray Metal